Calculation Results

COMPLETE

System: Two-zone momentary load demo

Voltage: 125.0 V

Highest peak on one DC supply: 25 A (DC 1)

Zone Summary

Stage totals are shown separately; stages and separate zone faults are not added together.

Zone	Per-stage total across assigned supplies	Status
T1 Transformer Differential (transformer_zone)	Primary trip: **Confirmed 39 A**	**COMPLETE**
BUS-1 Bus Differential (bus_zone)	Primary trip: **Confirmed 45 A** Reclose: **Confirmed 10 A** Motor inrush: **Confirmed 18 A** Motor run: **Confirmed 3 A**	**COMPLETE**

Warnings Summary

Zones requiring review

- None.

Missing or invalid currents

- None reported.

Other study warnings

- None.

Loads

Load	Quantity	Current per unit	Total current	Note
CB-HV - Trip Coil 1	1	10 A	10 A	breaker_1; Primary trip; DC 1
CB-HV - Trip Coil 2	1	8 A	8 A	breaker_1; Primary trip; DC 2
CB-LV - Trip Coil 1	1	12 A	12 A	breaker_2; Primary trip; DC 1
CB-LV - Trip Coil 2	1	9 A	9 A	breaker_2; Primary trip; DC 2
CB-LV - Trip Coil 1	1	12 A	12 A	breaker_2; Primary trip; DC 1
CB-LV - Trip Coil 2	1	9 A	9 A	breaker_2; Primary trip; DC 2
CB-F1 - Trip Coil 1	1	6 A	6 A	breaker_3; Primary trip; DC 1
CB-F1 - Trip Coil 2	1	5 A	5 A	breaker_3; Primary trip; DC 2
CB-F2 - Trip Coil 1	1	7 A	7 A	breaker_4; Primary trip; DC 1
CB-F2 - Trip Coil 2	1	6 A	6 A	breaker_4; Primary trip; DC 2
CB-F1 - Close Coil	1	10 A	10 A	breaker_3; Reclose; DC 1
CB-LV - Motor Inrush	1	18 A	18 A	breaker_2; Motor inrush; DC 1
CB-LV - Motor Run	1	3 A	3 A	breaker_2; Motor run; DC 1

Calculation Steps

1. Group actions by zone and operating stage.
2. Count each physical coil or motor once per stage; multiply entered current by the verified unit count.
3. Sum currents per DC supply within each stage, then take each supply's maximum across complete zones.
4. Stages and separate zone faults are not added together. Flagged zones have provisional results only.

Evaluation Output

```
Momentary DC protection-zone study: Two-zone momentary load demo
COMPLETE
Each configured zone is a separate fault event. Same-stage loads are
simultaneous; other stages are separate.
DC supplies are reported separately. No battery amp-hour calculation.
```

T1 Transformer Differential `(ct_1, ct_2, xfmr_2w_1) (Transformer)`

```
  CT boundaries: ct_1, ct_2
  Equipment: ct_1, ct_2, xfmr_2w_1
  Primary trip: 39 A across assigned supplies
    DC 1: 22 A
    DC 2: 17 A
    CB-HV - Trip Coil 1: 10 A × 1 = 10 A (DC 1)
    CB-HV - Trip Coil 2: 8 A × 1 = 8 A (DC 2)
    CB-LV - Trip Coil 1: 12 A × 1 = 12 A (DC 1)
    CB-LV - Trip Coil 2: 9 A × 1 = 9 A (DC 2)
```

BUS-1 Bus Differential `(bus_1, ct_3, ct_4, ct_5) (Bus)`

```
  CT boundaries: ct_3, ct_4, ct_5
  Equipment: bus_1, ct_3, ct_4, ct_5
  Primary trip: 45 A across assigned supplies
```

DC 1: 25 A
DC 2: 20 A
CB-LV - Trip Coil 1: 12 A × 1 = 12 A (DC 1)
CB-LV - Trip Coil 2: 9 A × 1 = 9 A (DC 2)
CB-F1 - Trip Coil 1: 6 A × 1 = 6 A (DC 1)
CB-F1 - Trip Coil 2: 5 A × 1 = 5 A (DC 2)
CB-F2 - Trip Coil 1: 7 A × 1 = 7 A (DC 1)
CB-F2 - Trip Coil 2: 6 A × 1 = 6 A (DC 2)
Reclose: 10 A across assigned supplies
DC 1: 10 A
CB-F1 - Close Coil: 10 A × 1 = 10 A (DC 1)
Motor inrush: 18 A across assigned supplies
DC 1: 18 A
CB-LV - Motor Inrush: 18 A × 1 = 18 A (DC 1)
Motor run: 3 A across assigned supplies
DC 1: 3 A
CB-LV - Motor Run: 3 A × 1 = 3 A (DC 1)

Peak per supply among confirmed, complete configured zones:
DC 1: 25 A — **BUS-1 Bus Differential** / Primary trip
DC 2: 20 A — **BUS-1 Bus Differential** / Primary trip

Appendix: Detected Topology Regions

Topology detection is retained for reference and is not automatically added to the protection calculation.

Region	Relationship	Conductors
Primary region: CB-F2, FEEDER 2 (off-sheet)	Enclosed	2
Primary region: CB-F1, FEEDER 1 (off-sheet)	Enclosed	2
Primary region: CB-HV, SOURCE (off-sheet)	Enclosed	2
Primary region: T1	Enclosed	2
Primary region: CB-LV	Enclosed	2
Primary region: BUS-1	Enclosed	3
Composite region: CB-LV, BUS-1, CT-BUS-IN	Outermost	5
Composite region: CB-HV, CT-HV, SOURCE (off-sheet) ...	Outermost	4
Composite region: CB-F1, BUS-1, CT-F1 ...	Outermost	5
Composite region: CB-F2, BUS-1, CT-F2 ...	Outermost	5
Composite region: CB-LV, CT-LV, T1	Outermost	4